

IAN SEQUEIRA

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EDUCATION

University of California, Santa Barbara

Santa Barbara, CA

B.S. Statistics & Data Science, B.A. Economics

Sept 2023 – June 2027

- GPA: 3.83 / 4.0 | Dean's Honors (7 quarters)
- Coursework: ML, Design of Experiments, Stochastic Processes, Regression Analysis, Econometrics

RESEARCH EXPERIENCE

UCSB Department of Statistics & Applied Probability

Santa Barbara, CA

Undergraduate Researcher – Prof. Annie Qu

Apr 2026 – Present

- Developing **Compositional Coordination Confidence (CCC)** for failure detection in multi-agent LLM pipelines, with anytime valid guarantees via typed e-processes
- Proving composition theorems; Monte Carlo simulations validated the framework across sequential, fan-in, and fan-out agent topologies

Irvine Summer Institute for Biostatistics & Undergraduate Data Science (ISI-BUDS)

Irvine, CA

Research Fellow – Prof. Dan Gillen & Dr. Josh Grill

June 2025 – Aug 2025

- Constructed GLM, ridge, and CART classifiers in R identifying Alzheimer's biomarkers (pTau217, PACC, ADL) across 1,150+ A4 trial participants (0.835 AUC); sample-size framework achieved 80% power with 188 vs. 500+ participants
- Co-author on biomarker-guided trial design study; research presented at *CTAD 2025*, San Diego; manuscript in preparation

WORK EXPERIENCE

Alida Inc.

Remote

Data Engineering Intern

Sept 2024 – Present

- Architected multi-mode AI data exploration platform on Databricks over 47 healthcare datasets using multi-agent routing, graph analysis, and automated SQL generation; extended with spec-driven ML prediction pipeline featuring typed contracts and dual-layer verification

Turing

Remote

Applied AI/GenAI SWE Intern

Feb 2026 – May 2026

- Delivered full SDLC on a 13-node LangGraph call triage pipeline (Intake Agent) through a 12-week industry sprint via ACM Industry at UCSB

PROJECTS

Healthcare Financial Toxicity Prediction

Mar 2026

R, XGBoost, BART, SHAP, fairmodels

- PSTAT 231 (graduate Statistical ML) final project on catastrophic out-of-pocket burden in 20,962 MEPS respondents; tuned 6 classifiers under 10-fold CV, XGBoost reaching ROC-AUC 0.874
- Evaluated subgroup fairness across race/ethnicity using DALEX, interpreting feature contributions as SHAP odds ratios

Research Advisor Finder

Nov 2025 – Feb 2026

FastAPI, PostgreSQL, pgvector, Next.js, OpenAI, Claude API

- Built full stack faculty discovery platform matching student research interests to 15k+ professors across 12 STEM fields via hybrid semantic search and CV-driven interest extraction

LEADERSHIP AND ACTIVITIES

Project Executive, ACM Industry, UCSB

Sept 2025 – Present

- Leading cross-functional student teams delivering applied data science and software solutions for industry partners

Finance & Funding Chair, CampMed, UCSB

Sept 2025 – Present

- Leading budget planning and sponsorship strategy for health equity outreach to underrepresented high school students

TECHNICAL SKILLS

Programming: R, Python, SQL, C++, JavaScript/TypeScript, HTML & CSS, LaTeX

Frameworks: tidymodels, xgboost, rstan, scikit-learn, PyTorch, SHAP, Databricks, LangGraph

Languages: English, Spanish | **Memberships:** ASA, IMS, Innovators Circle